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EnviroBot: A Mobile Robotic Arm that Detects and Picks Up Waste

Design, YOLOv8 Waste Detection and Prototype

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Jewel Nath: ORCID 0009-0002-6609-6378 · Code: https://github.com/jewel-nath/envirobot

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Abstract

EnviroBot is a mobile robot built to find waste on the floor and pick it up. It was built in 2023–24 at Comilla University, Bangladesh, by a team of three students: a four-wheel rover on a stainless steel base carries a six-servo aluminium arm with a gripper, a bin, and a camera.

The robot splits its work between two controllers. In its design, a Raspberry Pi 3 B handles vision; an Arduino Uno drives the arm through a PCA9685 servo driver and the wheels through two BTS7960 motor drivers, and takes commands over a serial link from the Pi and over Bluetooth from a phone app. Waste is found by a YOLOv8 model trained on 1,700 labelled images in three classes, plastic, paper and trash. The arm is modelled in URDF for ROS 2, and the design plans its motion with MoveIt.

In the team's evaluation, the model labelled 91% of paper, 94% of plastic and 92% of trash objects correctly; when it reported an object where there was none, it most often called it trash. On video, the detector, run on a laptop, labels waste in a live camera view, and the robot's arm picks objects up off the floor. This report describes the robot's mechanical build, electronics, control, arm model and waste detection, presents the evaluation, and sets out the lessons and the next steps toward collecting waste on its own.

Keywords: mobile manipulation, waste detection, YOLOv8, robotic arm, ROS 2, MoveIt, Arduino, Raspberry Pi, service robots.

1 Introduction

Litter on the floors of schools, offices and public places is still picked up by hand. Robots that find litter with a camera and pick it up have been shown in research [1], and modern object detectors such as YOLO [2] make the finding part fast enough to run in real time. Building such a robot, however, means joining several pieces that are usually studied apart: a mobile base, an arm, a detector, and the control that ties them together.

EnviroBot is a student team's answer to that problem at a low cost. Its aim was a robot that finds waste, labels what kind it is, and picks it up with an arm into a bin it carries (Fig. 1). The team chose off-the-shelf parts available in Bangladesh: an aluminium arm kit, hobby gear motors, a Raspberry Pi and an Arduino.

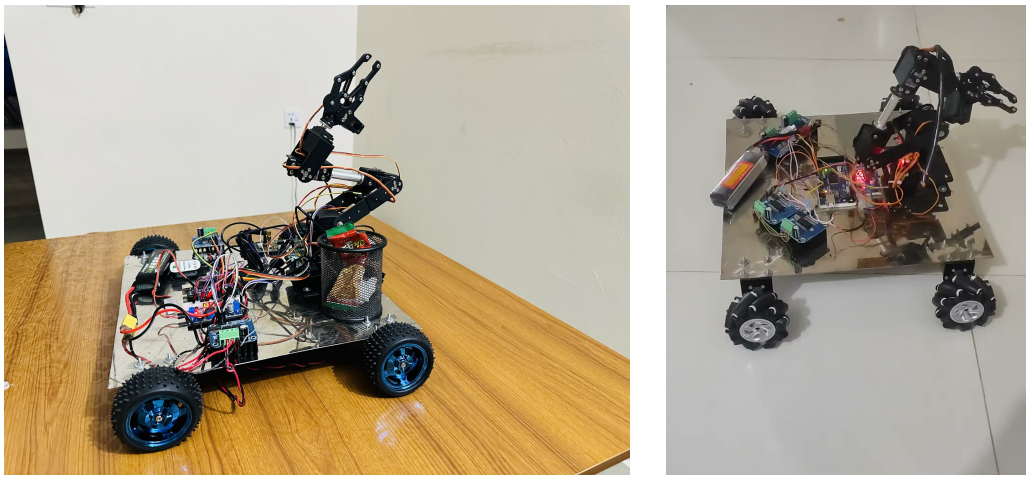


Figure 1: EnviroBot, finished, on 11 January 2024: the arm raised over the mesh bin, the electronics on the steel base (left). In a later driving test, on 11 May 2024, the arm rides on a base with mecanum wheels (right).

This report describes EnviroBot as built, and contributes:

1. a mobile manipulator built from low-cost parts: a 14-inch steel rover with four gear motors carrying a six-servo arm and a bin (Section 4);
2. a two-controller design that keeps vision on a Raspberry Pi and all motion on an Arduino, joined by a serial link, with manual control from a phone app over Bluetooth (Sections 5–6);
3. the arm modelled in URDF for ROS 2, moved joint by joint in RViz, with motion planning by MoveIt in the design (Section 7);
4. a YOLOv8 waste detector trained on 1,700 images in three classes, and its evaluation (Sections 8–9);
5. the behaviours shown on video, and the open source model of the robot, under the MIT licence [3] (Section 10).

Results at a glance

What EnviroBot does	How	Evidence
Labels waste by type	YOLOv8: 91% of paper, 94% of plastic and 92% of trash objects labelled correctly	measured
Finds waste in a live camera view	the detector boxes and names objects as they come into view	shown on video
Picks objects off the floor	the six-servo arm reaches down, grips and lifts	shown on video
Drives and moves its arm from a phone	an MIT App Inventor app over Bluetooth	built
Models the arm for ROS 2	the arm's URDF model, moved joint by joint in RViz	built
Drives on mecanum wheels	a later driving test of the rover with its arm	shown on video

The evaluation is the team's own, from their report of 14 January 2024. The videos were recorded on 11 January and 11 May 2024.

2 Related Work

Robots that collect litter. Bai et al. built a robot that finds garbage on grass with a deep network and picks it up with an arm [1]. EnviroBot follows the same pattern, a detector on board a mobile base that guides an arm, on indoor floors and with low-cost parts.

Detecting waste. YOLO frames detection as one pass of a network over the image [2]; YOLOv8 [4] is the version EnviroBot uses. General detectors are trained on datasets such as COCO [5]; for waste, TrashNet [6] classifies photos of single items into six classes, and TACO [7] annotates litter in the wild. EnviroBot's dataset, like TrashNet, has a few material classes.

Arms on ROS. ROS 2 [8] is the common middleware for robots today, and MoveIt [9] its motion-planning framework. Megalingam et al. modelled and controlled a 6-DOF arm in ROS [10], the approach EnviroBot takes for its arm.

3 System Overview

Fig. 2 shows the whole robot. Two controllers share the work. In the design, the Raspberry Pi is the vision computer: it reads the camera and runs the detector (in the recorded test, the detector ran on a laptop). The Arduino Uno is the motion controller: it sets the six arm servos through a PCA9685 servo driver [11] over I^2C, and the four wheel motors through two BTS7960 motor drivers [12]. The Pi and the Arduino talk over a serial link; a phone app sends the Arduino commands over Bluetooth. A 3-cell LiPo battery powers the robot.

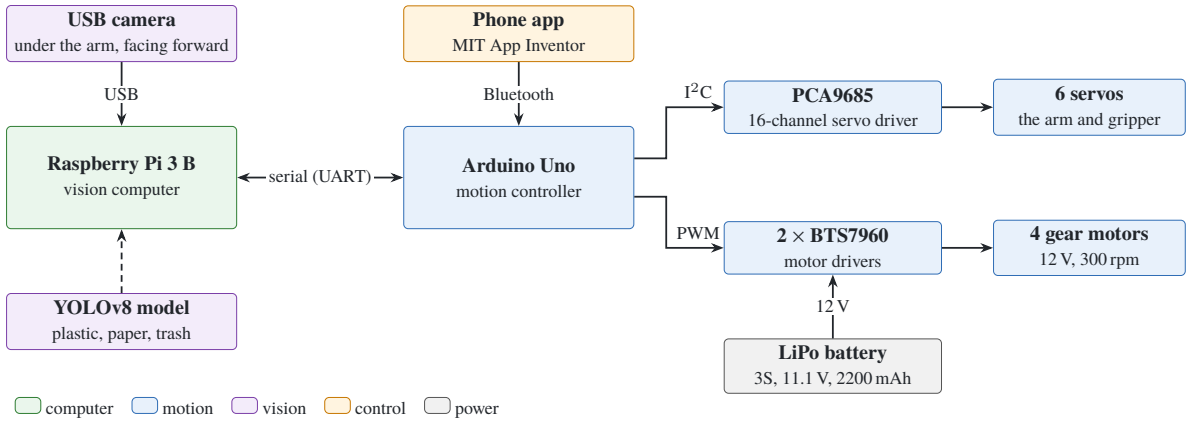


Figure 2: System overview. The Raspberry Pi (green) handles vision (purple); the Arduino drives the arm and the wheels (blue); the phone app (orange) gives manual control. From the team's circuit diagram and parts list.

The split follows from what each board does well. The Pi runs Linux, Python and the detector; the Arduino gives steady, timed output to the drivers, and lets the phone drive the robot without the Pi.

4 Mechanical Build

The rover. The arm is heavy for a small robot, so the team built a sturdy base: a 14 by 14 inch (356 mm) stainless steel sheet with a 12 V gear motor at each corner, each rated at 300 rpm with a torque sold as "10 kg" (Figs. 3 and 4). Each of the two motor drivers runs a pair of the motors.

The arm. The arm is an aluminium six-servo kit with a clamp claw: a waist, a shoulder, an elbow, a wrist that pitches and rolls, and a gripper. It is fixed at the front of the base, so that it reaches the floor in front of the robot without the base in its way. A mesh bin sits on the base beside it, within the arm's reach.

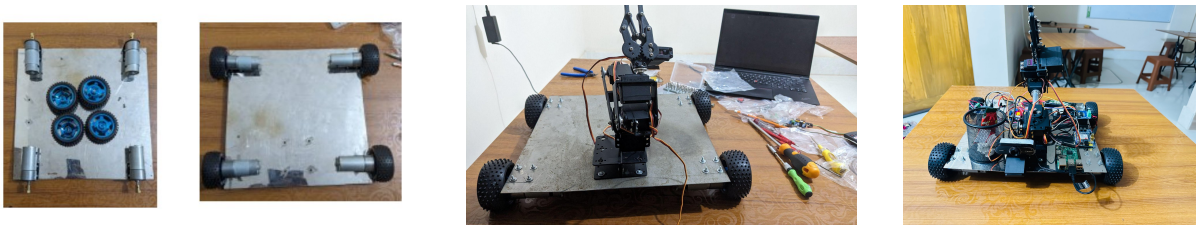


Figure 3: The build: the steel base with its four gear motors and wheels, from below and above (left); the arm fitted at the front of the base (middle); the finished robot with its bin, camera and boards (right).

Fig. 5 places the build in time, from the earliest dated image of the arm's model, in September 2023, to the finished robot in January 2024 and a later driving test on mecanum wheels in May 2024.

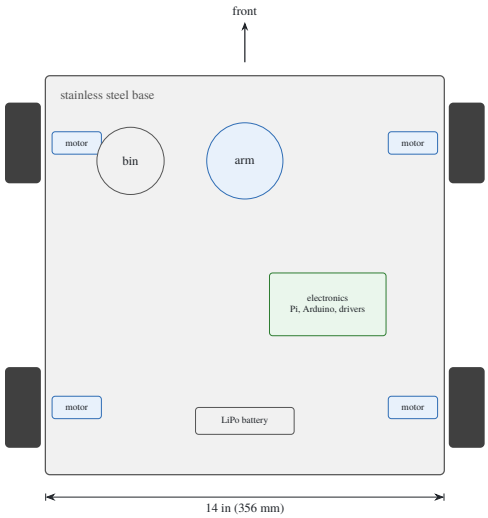


Figure 4: The rover from above, to scale for the base. The arm sits at the front; the positions of the bin, the electronics and the battery are drawn schematically.

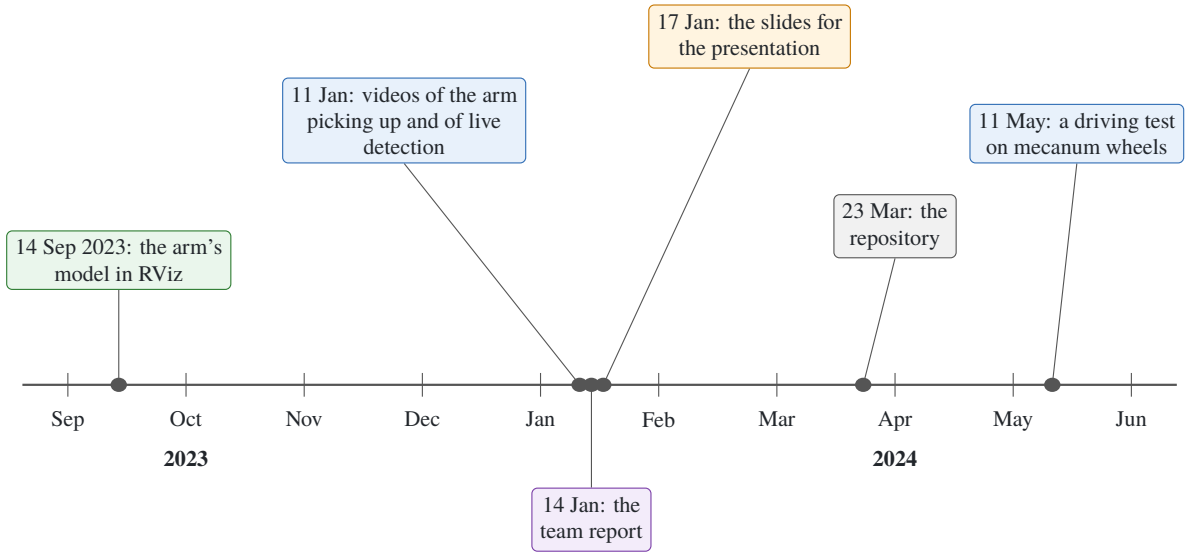


Figure 5: EnviroBot from the earliest dated image of the arm's model to the later driving test, from the dated photos, videos, report and repository.

5 Electronics

Table 1 lists the main parts, and Fig. 6 the wiring, from the team's circuit diagram of 14 January 2024. The Pi's UART (pins 8 and 10) is crossed to the Arduino's serial pins; the camera plugs into the Pi's USB port. The Arduino reaches the PCA9685 on its I^2C pins, A4 and A5, and sends PWM to the two BTS7960 drivers on pins 11 and 10 for the first and 6 and 5 for the second; each driver runs two motors. The drivers' enable pins are tied high, and their motor supply is the 12 V line from the 3-cell battery.

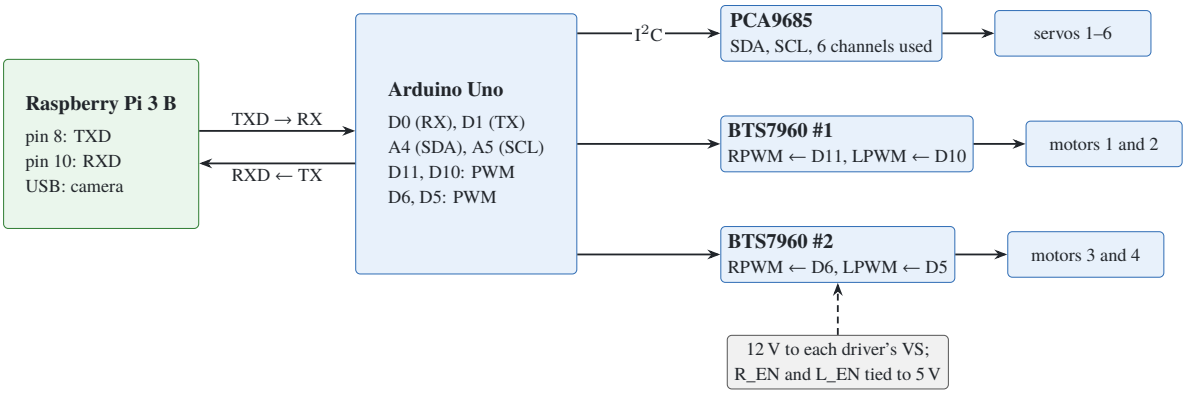


Figure 6: The wiring, pin by pin, from the team's circuit diagram (14 January 2024). The Bluetooth module is described in the report but not drawn in the diagram.

Table 1: Main parts, from the team report, the circuit diagram and the build photos.

Part	Role
Raspberry Pi 3 B	vision computer in the design: the camera and the detector
Arduino Uno	motion controller: the arm and the wheels; serial to the Pi, Bluetooth to the phone
PCA9685	16-channel, 12-bit PWM servo driver on I^2C; six channels used
6 servos	the arm kit's joints: waist, shoulder, elbow, wrist pitch, wrist roll, gripper
2 × BTS7960	H-bridge motor drivers, one per pair of wheels
4 gear motors	12 V, 300 rpm, torque sold as "10 kg", one per wheel
Camera	a USB webcam under the arm (the report also lists a Kinect sensor)
LiPo battery	3 cells, 11.1 V, 2200 mAh
Bluetooth module	on the Arduino, for the phone app

6 Control from a Phone

Before any autonomy, the team made the robot fully controllable by hand, which let them test every joint and the drive on their own. The app, built in MIT App Inventor [13], connects to the Arduino over Bluetooth (Fig. 7). Its arm screen has a pair of buttons for each joint, grip, wrist pitch, wrist roll, elbow, shoulder and waist, a slider for the arm's speed, and buttons to save a sequence of positions, run it back, and reset. Other buttons drive the rover.

The motors and servos were brought up step by step. Each motor was tested with its driver from the Arduino's serial monitor, and its wiring or code corrected until it turned the right way. Each servo was calibrated the same way: its starting, lowest and highest angles were found and written down, so the arm's commands stay within each joint's range.

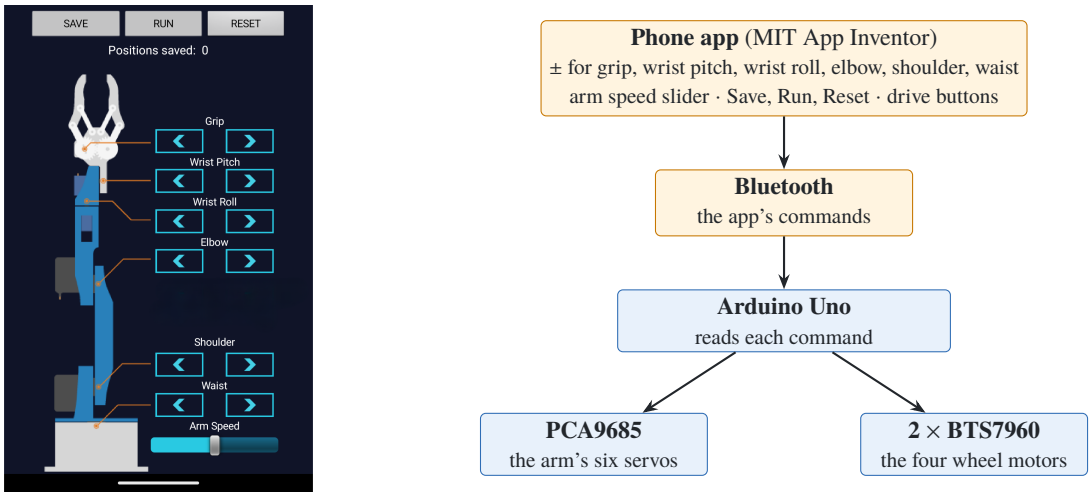


Figure 7: Manual control. The app's arm screen (left): a pair of buttons for each joint, a speed slider, and Save, Run and Reset. Each command goes over Bluetooth to the Arduino, which moves a joint through the servo driver or the wheels through the motor drivers (right).

7 Arm Model and Motion Planning

The arm is described in URDF, through xacro, as a ROS 2 package, `smart_arm` [3, 8]. Its chain has six controlled joints, from the waist to the gripper, plus a second finger joint that mirrors the first, so one servo opens and closes both fingers (Fig. 8). Each joint has limits on its travel. A second file models the rover base, with a continuous joint for each wheel.

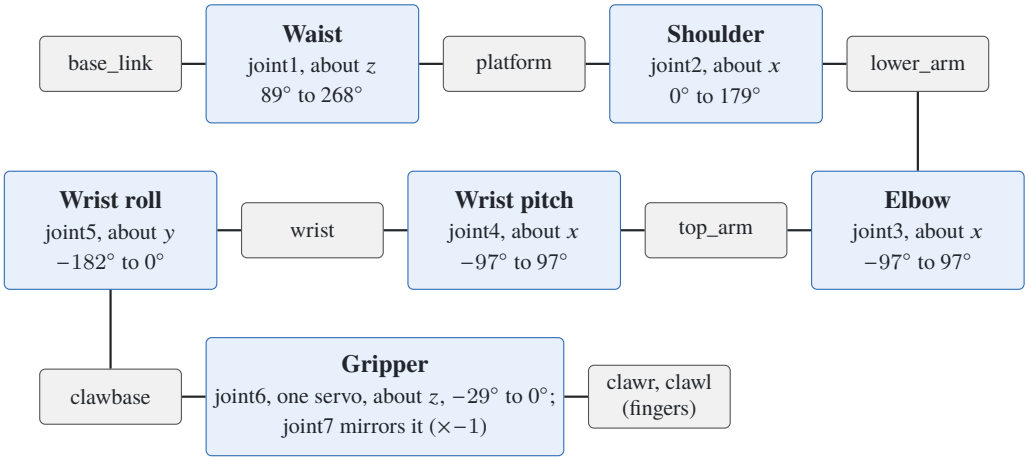


Figure 8: The arm's kinematic chain in the URDF model: links (grey) and joints (blue), each joint named as on the app's arm screen, with its axis and its range in degrees.

The team first moved the model joint by joint in RViz, with a slider for each joint, in September 2023. The team report describes the next step, setting the arm up in MoveIt [9], which plans a collision-free path for the arm to a goal pose (Fig. 9). A ros2_control set-up names the same six joints for a joint-trajectory controller; it is not yet wired into the model.

The team designed the full autonomous pick-up on this base (Fig. 10): a ROS 2 node runs the detector on each camera frame and passes on the class and position of each item; a target is chosen from it; MoveIt plans the arm's path to it; and the Arduino moves the joints along the plan. Each part was shown on its own, as Sections 9 and 10 describe; joining them into one program is the next step (Section 12).

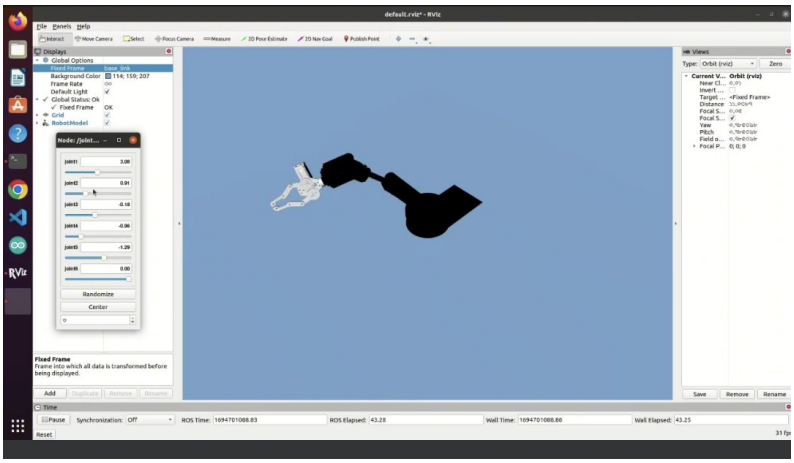
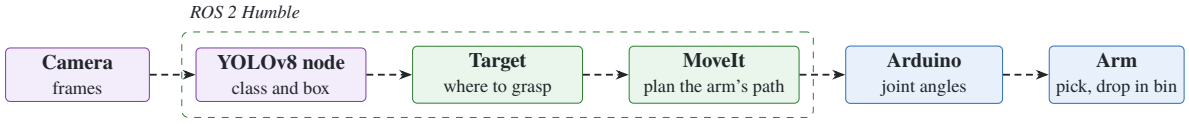


Figure 9: The arm's model in RViz, moved with a slider for each joint, on 14 September 2023.



Shown on their own: live detection (on a laptop), the arm's model in RViz, and the arm picking up an object.

Figure 10: The autonomous pick-up, as designed: from a camera frame to the arm, through the detector, a chosen target and MoveIt.

8 Waste Detection with YOLOv8

Data. The team's dataset holds 1,700 images in three classes, plastic, paper and trash, many of them from an earlier project in their department (Fig. 11). Each item in each image is labelled with a box and a class. The set was enlarged by changing the brightness of some images and shifting and rotating others, and every image was resized to 512 by 384 pixels with OpenCV. It was split 70/20/10 into training, validation and test sets.[1]

Model. The detector is YOLOv8 [4] in its large size, YOLOv8l, trained for 50 epochs in Google Colab. YOLOv8 finds the objects in an image and classifies them in one pass of the network [2]; its loss combines binary cross-entropy for the classes with distribution focal and CIoU losses for the boxes.

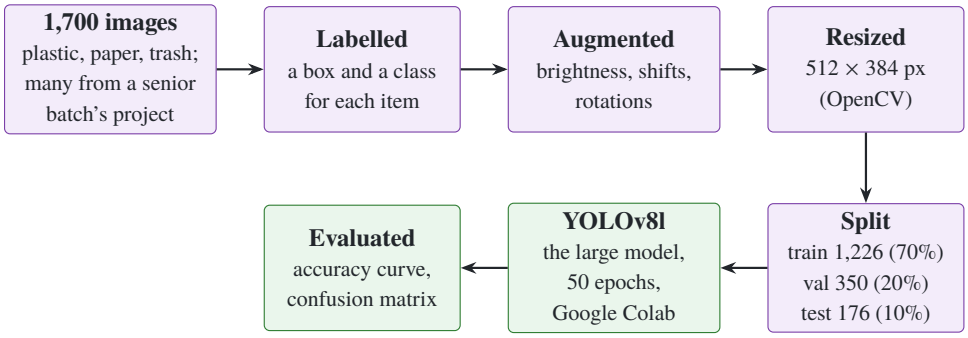


Figure 11: From images to a trained detector: labelling, augmentation, resizing, the split, and training YOLOv8l.

[1] The report gives the split as 1,226, 350 and 176 images, which add up to 1,752 rather than 1,700; the report does not explain the difference.

9 Evaluation

The team's report of 14 January 2024 gives the training curves and a confusion matrix; the figures below are typed in from it. The report does not say which split the matrix was computed on.

Training. Accuracy rose unevenly at first, with sharp drops at epochs 5–6, 19 and 30, then settled from epoch 34 and reached 0.81 by epoch 50; the loss fell from 2.57 to 0.37, with a clear step down after epoch 40 (Fig. 12). The report does not name the accuracy metric, so this curve is best read as the trend of training rather than a measure to compare with other detectors.

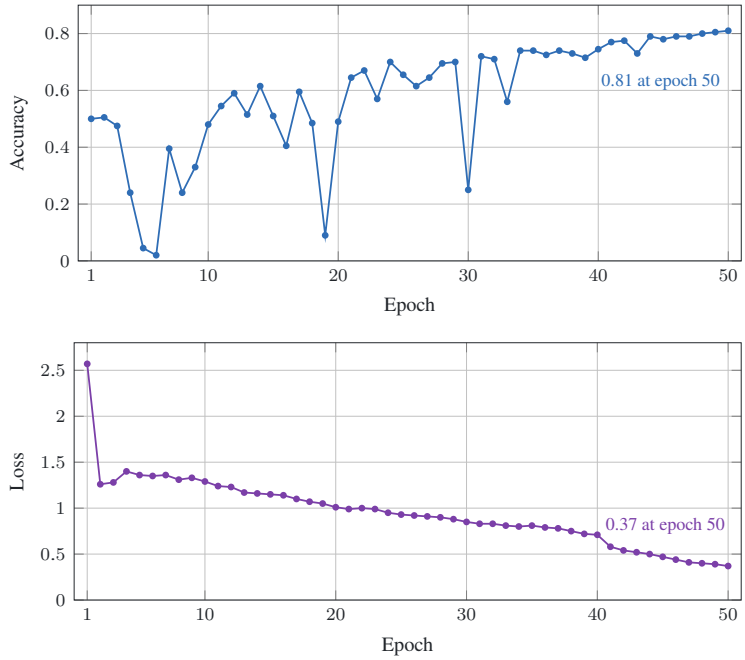


Figure 12: Training over 50 epochs: the accuracy the team reported (top) and the loss (bottom). Values read off the report's charts, to about ±0.02.

Per class. The confusion matrix (Fig. 13) shows, for the objects of each true class, the share given each label. The model labelled 91% of paper objects, 94% of plastic and 92% of trash correctly; confusions between the three classes were at most 5%, and a few paper and trash objects (3% and 1%) were missed as background.

The background column tells the rest of the story. When the model reported an object where there was none, it called it trash half the time (0.50), plastic 30% and paper 20%. Trash, the broadest of the three classes, drew the most.

Key finding

The detector tells waste types apart well: 91–94% of the objects in each class were labelled correctly, and no two classes were confused more than 5% of the time. When it reported an object where there was none, it most often called it trash; more background images in training might reduce this.

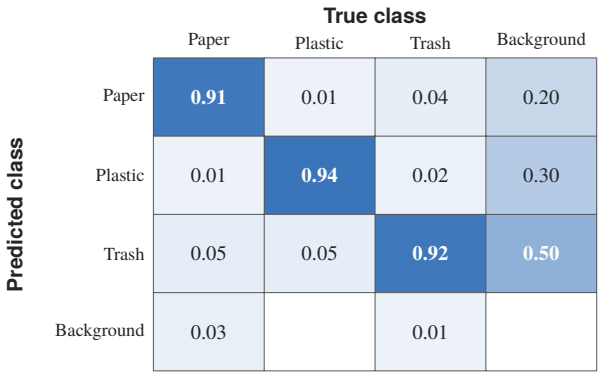


Figure 13: The detector's confusion matrix, from the team report. Each column is one true class and sums to 1, to rounding; the diagonal is the share labelled correctly. The background column shows what the model's false detections were labelled.

10 Demonstrations

The robot's abilities were recorded on video (Table 2, Fig. 14).

Table 2: Behaviours recorded on video.

Date	What the video shows
11 Jan 2024	The detector running live on a laptop, boxing and naming objects as they come into view (42 s)
11 Jan 2024	The arm reaching down from the rover, gripping a small cube off the floor and lifting it (73 s)
11 Jan 2024	An arm test on the floor: the arm reaching down to a small box on the floor (77 s)
11 Jan 2024	Close-ups of the finished robot: the arm, the bin, the battery and the boards (20 s)
11 May 2024	A driving test of the rover with its arm, on mecanum wheels (16 s)

The live test used a model that also names metal, which the three-class model in the report does not, so the live test ran a different version of the detector. The robot was presented at Comilla University in January 2024.

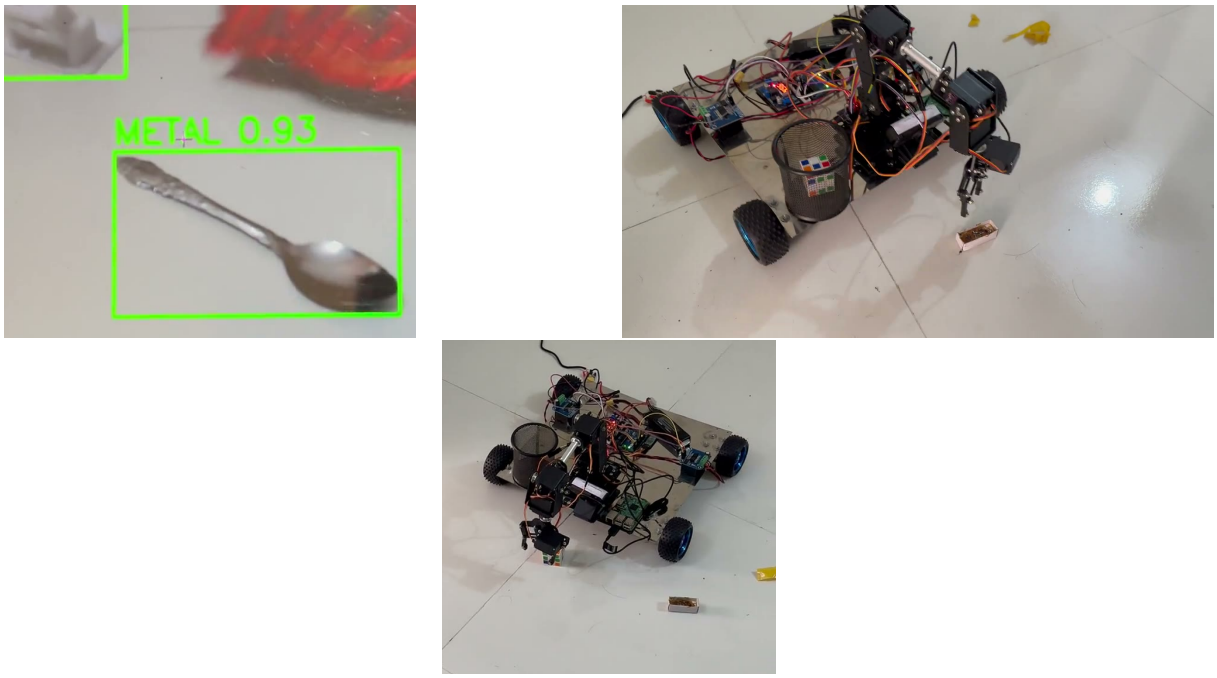


Figure 14: From the videos of 11 January 2024: the live detector boxing a spoon and naming it metal (left); the arm reaching for a box on the floor (middle); the gripper holding a cube it picked off the floor (right). In both arm videos the rover is on a power cable.

11 Lessons Learned

Build the base for the arm. The arm, not the rover, set the design: the team chose a steel base and four geared motors to carry it, at the cost of a heavier robot.

Split the work between two boards. Keeping all motion on the Arduino meant the arm and wheels could be driven from the phone on their own.

Manual control first. The phone app let the team test and calibrate every joint and the drive before any autonomy.

Reuse data, and name the metric. Starting from an earlier project's images and enlarging them with augmentation gave a usable detector. Recording which metric each number is, and keeping a set of background images, would have made the evaluation stronger.

12 Limitations and Future Work

EnviroBot's detector was evaluated by the team in its report; its pick-ups and driving were recorded on video but not measured. The detector ran on a laptop in the recorded test, and the pick-up and the detector were shown separately. The repository holds the robot's model and ROS 2 package, while the detector, the Arduino code and the app stayed with the team.

Open steps follow from these (Fig. 15): one program from the camera to the grasp, running the detector on the robot's own computer, and a measured pick-up success rate on a fixed set of objects; then driving a path and collecting on its own, and sorting waste into separate bins by class. The team's report also proposed Lidar for mapping and navigation, a depth camera to locate waste more precisely, and an industrial-grade arm.

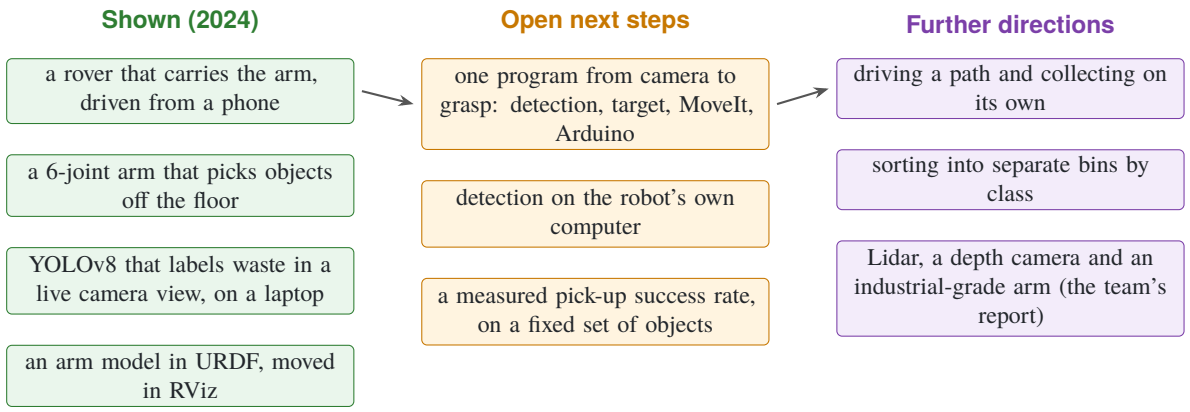


Figure 15: What EnviroBot showed in 2024, and the open steps.

13 Conclusion

EnviroBot shows that a small team can build a mobile robot that finds waste and picks it up from low-cost parts: a steel rover, a six-servo arm, a Raspberry Pi and an Arduino. Its YOLOv8 detector labelled plastic, paper and trash correctly more than nine times in ten in the team's evaluation, its arm picks objects off the floor, and the whole robot can be driven from a phone. Its model and ROS 2 package are open for others to build on.

Author Contributions

All three authors designed, built and tested EnviroBot together at Comilla University. Jewel Nath put the robot's model into its public repository and wrote this report.

Acknowledgements

The authors thank their supervisor, Amena Begum, Assistant Professor in the Department of Information and Communication Technology at Comilla University, for her guidance throughout the project.

The detector is built on Ultralytics YOLOv8 (AGPL-3.0). The ROS 2 package's test files come from the ROS 2 package template and keep the Open Source Robotics Foundation's Apache 2.0 notices.

This report was written with the help of AI assistants, under the first author's direction; its statements were

checked against the team's report, the circuit diagram, the repository and the dated photos and videos.

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A Additional Material

Table 3: The repository.

Folder	Holds
`src/smart_arm/description/`	the rover base and the arm in URDF/xacro, with their meshes
`src/smart_arm/launch/`	a launch file that shows the rover base in RViz with a slider for each wheel; the arm's include is commented out
`src/smart_arm/config/`	a ros2_control controller set-up for the arm's six joints, not yet wired in
`src/smart_arm/worlds/`	an empty Gazebo world
`docs/media/`	the photos and videos
`docs/report/`	this report, its figures and how to build it
`docs/diagrams/`	every diagram as its own PDF and PNG